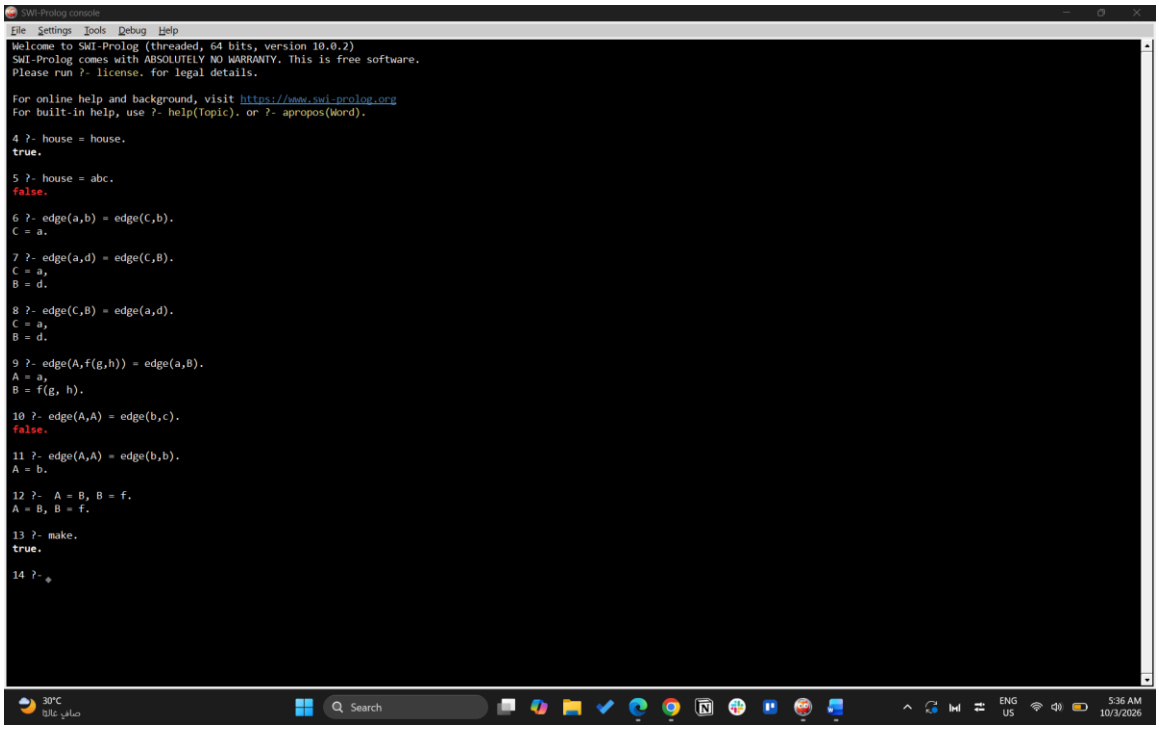
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ARTI401 | Artificial Intelligence: Principles and Techniques

Lab 4: Variables in Prolog

Task 1



Question 1

Query:?- house = house.

Result: true.

Explanation: Both terms are the same, so they can be unified.

Question 2

Query:?- house = abc.

Result: false.

Explanation: house and abc are different atoms, so they cannot be unified.

Question 3

Query:?- edge(a,b) = edge(C,b).

Result: C = a.

Explanation: Both terms have the same predicate name and number of arguments. C can be unified with a, while b matches b.

Question 4

Query:?- edge(a,d) = edge(C,B).

Result: C = a,

B = d.

Explanation: The predicate names and number of arguments are the same. C is unified with a and B is unified with d.

Question 5

Query:?- edge(C,B) = edge(a,d).

Result: C = a,

B = d.

Explanation: The variables C and B can be instantiated with a and d, respectively, so the two terms can be unified.

Question 6

Query:?- edge(A,f(g,h)) = edge(a,B).

Result: A = a,

B = f(g,h).

Explanation: A is unified with a, and B is unified with the term f(g,h). Therefore, the two terms can be unified.

Question 7

Query:?- edge(A,A) = edge(b,c).

Result: false.

Explanation: The same variable A appears twice, so both positions must have the same value. However, b and c are different, so the terms cannot be unified.

Question 8

Query:?- edge(A,A) = edge(b,b).

Result: A = b.

Explanation: The same variable A appears twice, and both positions contain b. Therefore, A can be unified with b.

Question 9

Query:?- A = B, B = f.

Result: A = f,

B = f.

Explanation: First, A and B are unified. Then B is unified with f, so both A and B have the value f.

Question 10

Query:?- A = B, B = f, A = e.

Result: false.

Explanation: First, A and B are unified, and B is then unified with f. Therefore, A must also be f. The final condition A = e conflicts with this, so the query fails.

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Task 2

